Dear Hiring Manager,

I am writing to express my interest in the Quantitative Researcher position at WorldQuant. As a recent MSc Computer Science graduate from Sapienza University of Rome, specializing in Artificial Intelligence and Machine Learning, I am excited by the opportunity to contribute to WorldQuant's research-driven approach to discovering predictive signals in global financial markets.

My academic and research experience has equipped me with a strong foundation in quantitative analysis, machine learning, statistical modeling, and experimental research. During my master's thesis, *Evaluating Pretrained Vision Transformers Under Continual Learning*, I designed reproducible research pipelines, benchmarked transformer-based models, and analyzed robustness, knowledge retention, and performance across sequential learning tasks using Python and PyTorch. This work required rigorous experimentation, critical analysis, and a systematic approach to evaluating complex models—skills that closely align with the research culture at WorldQuant.

Beyond my thesis, I have worked on projects involving transformer-based natural language processing, semantic embeddings, and machine learning model evaluation. My DocDelta project involved developing an AI-powered document intelligence platform using Sentence Transformers, FastAPI, Docker, and AWS, while my semantic text compression project explored transformer embeddings, quantization techniques, and quantitative evaluation using cosine similarity and ROUGE metrics. These experiences strengthened my ability to transform complex data into meaningful insights through careful experimentation and analytical reasoning.

I am particularly drawn to WorldQuant because of its emphasis on intellectual curiosity, independent research, and solving challenging problems through data. I enjoy exploring difficult questions, designing experiments, and continuously refining solutions based on empirical evidence. While my background is rooted in artificial intelligence and machine learning rather than finance, I am eager to apply my quantitative research skills to financial markets and learn from the outstanding researchers at WorldQuant. I view this opportunity as an exciting intersection of mathematics, programming, data analysis, and scientific discovery.

In addition to my research experience, I bring strong programming skills in Python, SQL, and C++, along with experience using PyTorch, TensorFlow, Scikit-learn, Linux, and statistical analysis techniques. My IEEE conference publication further reflects my ability to conduct structured research, communicate technical findings effectively, and contribute to solving complex analytical problems.

I am excited about the opportunity to contribute to WorldQuant's mission of developing innovative quantitative strategies through rigorous research and data-driven thinking. I would be grateful for the opportunity to discuss how my research background, technical skills, and enthusiasm for quantitative problem-solving can contribute to your team.

Thank you for your time and consideration. I look forward to hearing from you.

Sincerely,

Swati Aggrawal